

# **Data visualization** guide

How McKesson Workforce Innovation turns workforce data into charts that lead to decisions.

Version 1.1.0 October 2026 All example data is synthetic

OVERVIEW

# **Start here:** the quick path

You don't need all of it. These pages cover most of what goes into a deck; the rest is reference for when you build a particular chart.

On screen, each chart family shows its anatomy and best practice first; variants and build steps open with **More**. The short **Quick guide** PDF has just the pages below.

### 1 · READ ONCE (20 MINUTES)


- **Start with the question** 7
- **Anatomy of a chart** 9
- **Write the headline last** 10
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- **Honest scales** 15
- **People data is personal** 16


### 2 · BEFORE EVERY DECK


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- **Clarity first** 20
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### 3 · WHEN YOU BUILD


- **Bar chart, the default** 31
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- **Set up your tools once** 112
- **Cheat sheet** 110
- **Check before you send** 113


Any other chart: find it in the chooser, read its anatomy and best practice, and open **More** for the build steps.

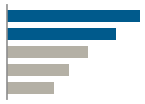
CHART CHOOSER

# **Chart** chooser

Pick the chart from the question, not from habit.

1. Write the question the chart answers in one sentence.
2. Find the question type and start with its first choice.
3. Check the traps on the next page.
4. Follow the chart's page to build it.

Every chart name links to its page in this guide. **Avoid** lists the charts that look right for the question but mislead or slow the reader.



## Magnitude

*How big? Which is bigger, by how much?*

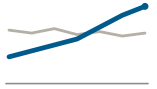
| | |
|---|---|
| START WITH | **Sorted bar** |
| ALSO | Dot plot for values far from zero; paired bars |
| AVOID | Rainbow, 3D or cut-off bars |



## Part-to-whole

*What is it made of? How does a total break down?*

| | |
|---|---|
| START WITH | **100% stacked bar** |
| ALSO | Waterfall between totals; treemap for many parts |
| AVOID | Pies over 5 slices |



## Change over time

*How is it moving? When did it change?*

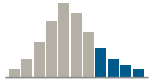
| | |
|---|---|
| START WITH | **Line**, focus series in blue |
| ALSO | Stacked area; small multiples; bubble timeline |
| AVOID | Ten colored lines |



## Ranking

*Who's on top? Who moved?*

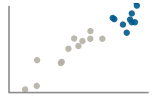
| | |
|---|---|
| START WITH | **Sorted bar**; **slope** for change |
| ALSO | Dumbbell for us vs. them; heat table |
| AVOID | A color for every line |



## Distribution

*How spread out? What's typical?*

| | |
|---|---|
| START WITH | **Strip plot** or **histogram** |
| ALSO | Box plot to compare many groups |
| AVOID | An average with no spread |



## Correlation

*Do two measures move together?*

| | |
|---|---|
| START WITH | **Scatter** or bubble scatter |
| ALSO | Heatmap: two categories, one measure |
| AVOID | Dual axes |



## Relationships and flows

*How do things connect or move?*

| | |
|---|---|
| START WITH | **Sankey** |
| ALSO | Parallel coordinates; Venn, 3 sets max |
| AVOID | A color per stage |



## Geo-spatial

*Where? Does place matter?*

| | |
|---|---|
| START WITH | **Choropleth** for rates |
| ALSO | Bubble map for counts; world map; zoom-in |
| AVOID | Counts on a choropleth |




## Single number

*What one figure should they remember?*

| | |
|---|---|
| START WITH | **Big number** with a comparison |
| ALSO | Progress bar or ring for a target |
| AVOID | Speedometers |

CHART CHOOSER

# **Before you build:** the common traps

The right chart type can still go wrong. If you are about to do one of these, use the fix instead.

The page numbers refer to this guide's PDF; on screen, each fix links to its page.

| ABOUT TO | INSTEAD | |
|---|---|---|
| Cut the bar axis so small differences show | Bars from zero; a dot plot for a narrow range | p. 33 |
| Give each bar its own color | Grey bars, the one that matters in blue | p. 33 |
| Put two units on two axes | Two aligned panels, or index both to 100 | p. 76 |
| Draw ten lines in ten colors | All grey, one or two in blue; or small multiples | p. 49 |
| Make a pie with eight slices | A sorted bar; donuts take five parts at most | p. 45 |
| Stack eight segments in a bar | Five at most, the story segment blue on the baseline | p. 38 |
| Color a waterfall green up, red down | Favorable green, unfavorable orange; or grey with one blue | p. 41 |
| Map raw counts by state | Map rates; show counts as bubbles | p. 94 |
| Show a KPI as a speedometer | A big number with a comparison, or a bar to target | p. 101 |
| Color workstreams red and green, and nothing else | Symbol and word with every dot; status colors only for status | p. 107 |
| Paste a 30-row table to make one point | Chart the point; keep the table for the appendix | p. 64 |
| Put four charts on one slide | Two at most, each with its own insight headline | p. 21 |
| Show a team of 7 people | Suppress or combine groups under 10 | p. 16 |

FUNDAMENTALS

# **Start with the question,** not the chart

A chart is finished when a reader can say what it means and what to do about it, without you in the room.

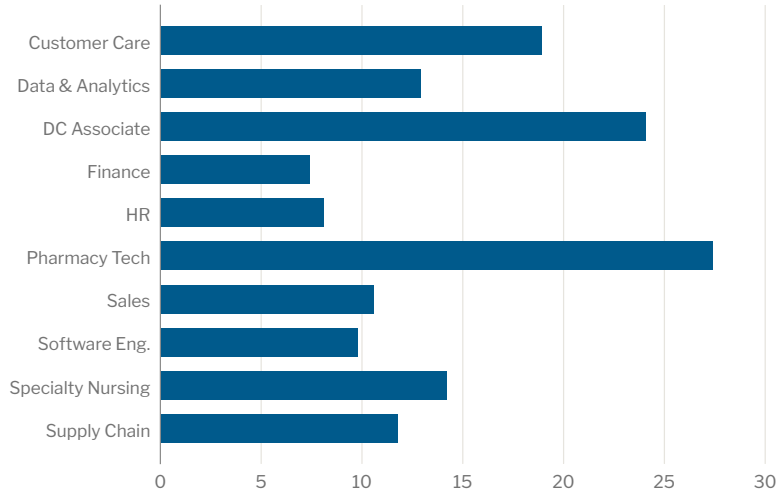
Before you open a tool, write the question in one sentence: "Where is attrition concentrated?", not "show attrition".

Then pick the mode. **Explanatory** charts prove a finding, for decks and reports (the default). **Exploratory** views let readers find answers, for dashboards. Both follow the same rules.

Both charts here are accurate; only one is useful in a meeting.

### Attrition by job family

FY26

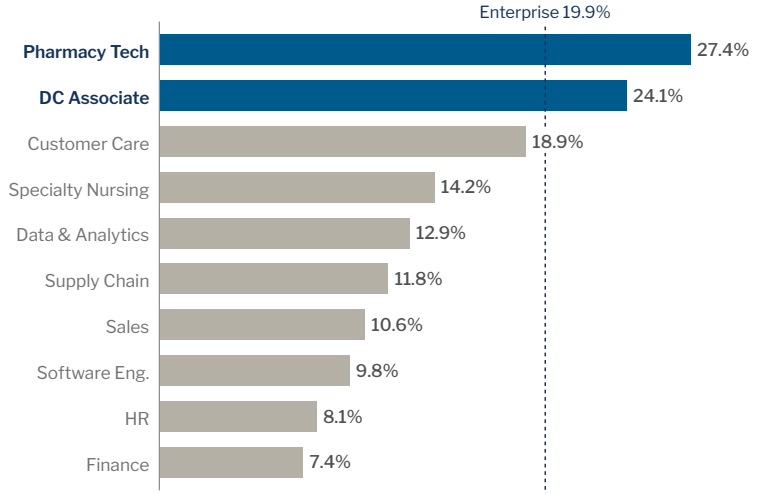



**Shown**

A label for a title, alphabetical order, every bar equal. The reader has to find the point alone.

### Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate

Voluntary attrition by job family, FY26 (%)




**Explained**

The headline states the finding, the order ranks the families, and color points to the two the headline names.

FUNDAMENTALS

# **Anatomy** of a chart

Every McKesson chart has the same four layers, top to bottom: headline, subtitle, chart, and footnote with source.

The headline and subtitle do different jobs. The headline is editorial: it states the finding. The subtitle is neutral: metric, unit, scope and period.

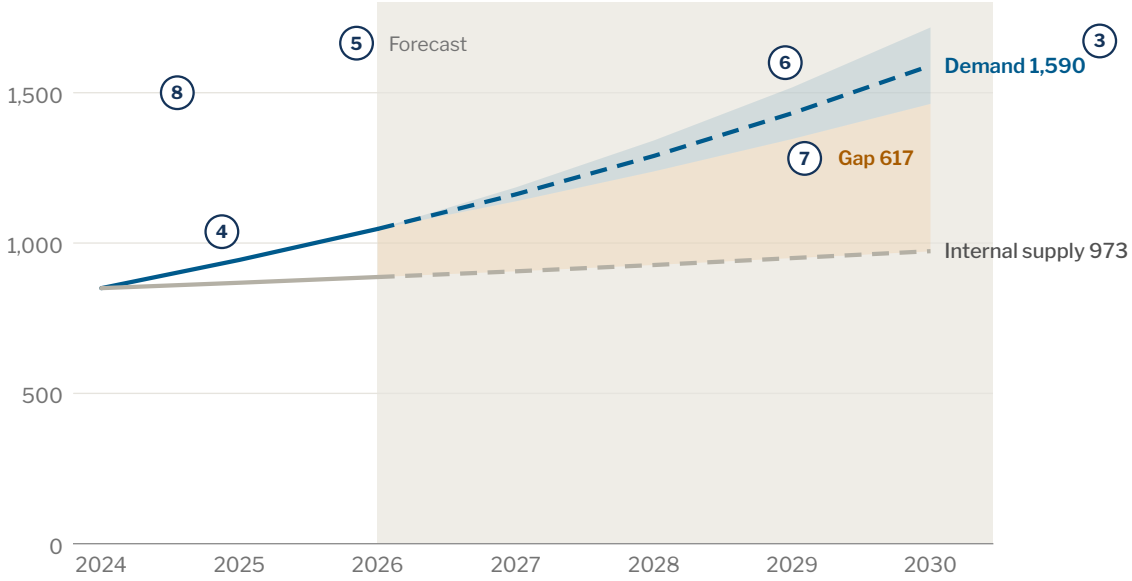
Inside the chart, everything is labelled in place, so the reader never has to match colors to a legend.

All four layers share one left edge. In HTML they are page text; the chart library draws only the data.

1. **Insight headline**
   A full sentence with the number. At most two lines, no final period.
2. **Subtitle**
   Metric, unit, scope and period. No opinions.
3. **Direct labels**
   Series named at the line end, in the series color. No legend.
4. **Focus and context**
   The series the headline is about is blue; the comparison is grey.
5. **Forecast marked**
   A shaded period and dashed lines separate estimates from actuals.
6. **Range, not a point**
   Forecasts show their uncertainty.
7. **The point, annotated**
   The gap the headline quotes, labelled where it is.
8. **Light scaffolding**
   Few, light gridlines; units live in the subtitle.
9. **Footnote and source**
   Definitions, system, as-of date, population.

## Internal supply covers only 61% of Data & AI demand by 2030, a gap of about 620 FTE (1)

Data & AI roles: forecast demand vs. internal supply, 2024–2030 (FTE) (2)




Forecast range on demand widens to ±8% by 2030. Internal supply = current staff plus planned reskilling, less expected attrition.

Source: Workday HCM and SWP model (synthetic data for illustration), as of 31 Mar 2026 (9)

FUNDAMENTALS

# **Write the headline** last, from the numbers

The headline is the most-read text on the page. It should be the finding, not the topic.

### The formula

**Who or what + verb + number + so what.**

### Before you keep it

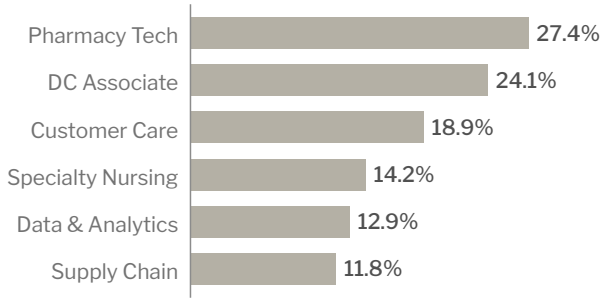
- [ ] Recompute every number, "all", "most" and "n of m" from the data
- [ ] The blue in the chart is what the headline names
- [ ] Two lines at most; no final period

If you can't write a headline, you don't have a finding yet: you have a chart.

| | |
|---|---|
| **LABEL** | Attrition by job family · names the topic; says nothing |
| **VAGUE** | Attrition varies across job families · true of every chart ever made |
| **OPINION** | Attrition is out of control in the frontline · a claim the chart can't support |
| **INSIGHT** | **Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate** |
| **INSIGHT + ACTION** | **Two frontline families account for 73% of voluntary exits; retention there moves the enterprise rate** |

## Attrition varies across job families

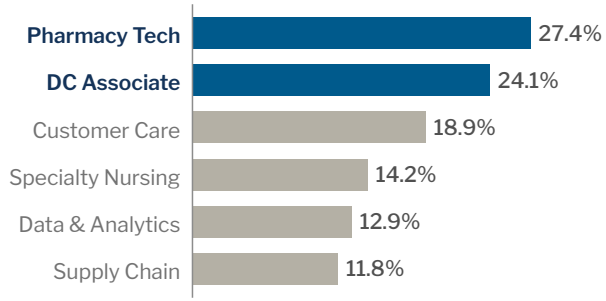
Voluntary attrition by job family, FY26 (%)




**Headline and chart both neutral**
Nothing tells the reader where to look.

## Pharmacy and DC teams lose about 1 in 4 people a year

Voluntary attrition by job family, FY26 (%)




**Headline and highlight agree**
The two blue bars are exactly what the sentence names.

FUNDAMENTALS

# **Grey first,** then color the point

Color is for meaning, never decoration. Start with every mark in grey, then color only what the headline names.

Highlight at most two items. When everything is highlighted, nothing is.

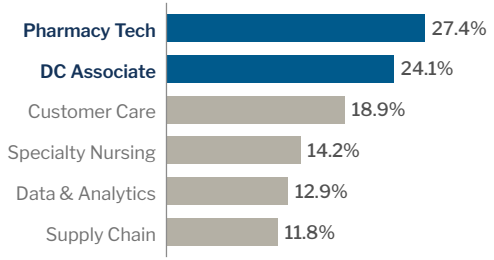
Echo the highlight in the text: bold the matching category label, and set callouts in the color of the mark they describe.

**Orange means "worse".** Keep it for negatives: below target, shrinking, a shortfall. Never use it as a plain highlight; readers take it as a warning.

PATTERN 1 · DEFAULT

### Focus

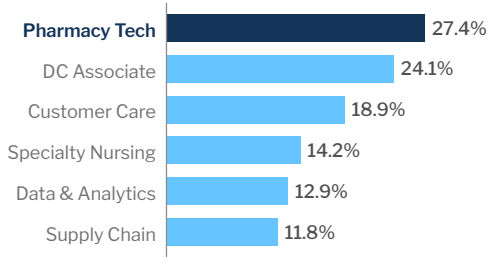
All grey; the items the headline names in blue.




PATTERN 2

### Uniform + accent

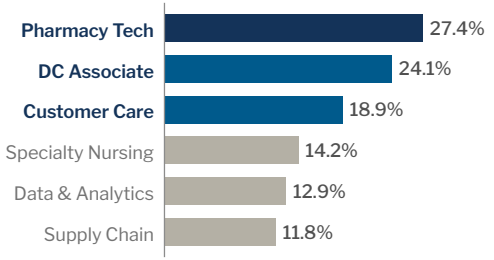
One series in light blue; the single exception in navy.




PATTERN 3

### Focus + accent

Grey context, a blue group, one navy exception. Use sparingly.




### Color roles

| | | |
|---|---|---|
| **Context** · everything the reader should see but not focus on | `#B4B0A5` |
| **Focus** · what the headline names | `#005A8C` |
| **Accent** · the one exception that must pop (navy, not orange) | `#143359` |
| **Light blue** · the series when navy marks the exception | `#66C4FF` |
| **Orange** · only when it means worse: below target, shrinking, a shortfall | `#EF8200` |
| **Green** · only when it means better, with ▲ or words | `#00AF68` |
| **Total and benchmark** · totals, targets, reference lines | `#143359` |
| **Context light** · de-emphasized fills in stacks and areas | `#DEDBD1` |

FUNDAMENTALS

# **Honest** scales

A chart can be accurate in every number and still mislead. These four rules prevent the common ways.

**Bars start at zero.** If the differences only show on a narrow range, change the chart, not the axis.

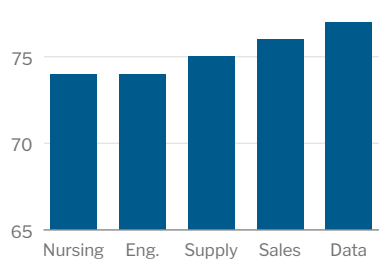
**No dual axes.** Two scales let you make any two lines cross wherever you like.

**Bubble area, not diameter,** is proportional to value.

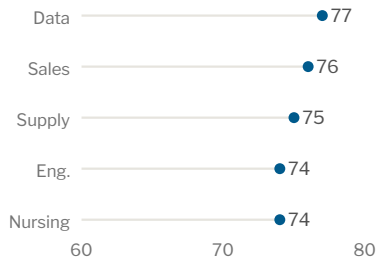
**Pies take five slices at most,** and sum to 100%.

The one exception: a waterfall bridging two large totals may break its axis, if the break is drawn and footnoted.

## Truncated bars

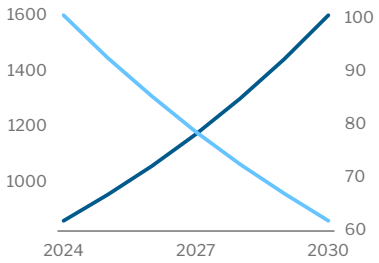



**Try to avoid**
An axis from 65 turns a 4% difference into a 33% one.

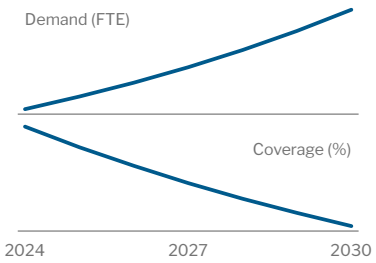



**Recommended**
A dot plot compares position, so the range can be narrow.

## Dual axes

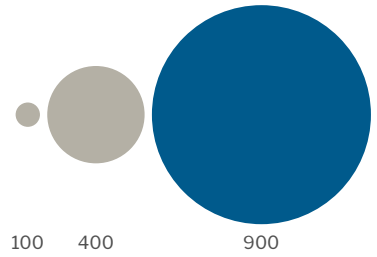



**Try to avoid**
The crossing point is an artifact of two chosen scales.

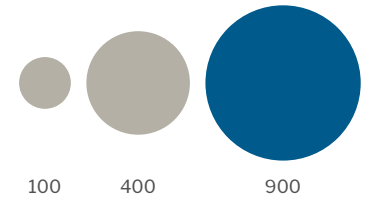



**Recommended**
Two aligned charts share the time axis.

## Bubble size

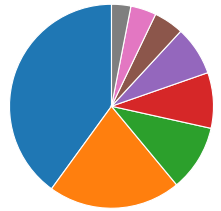



**Try to avoid**
Sized by diameter, 900 looks 81× bigger than 100.

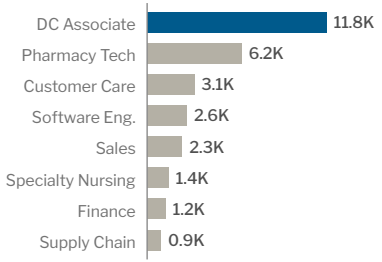



**Recommended**
Sized by area, 900 is 9× bigger, as it should be.

## Too many slices



**Try to avoid**
Eight similar angles can't be ranked by eye.




**Recommended**
A sorted bar ranks them instantly.

FUNDAMENTALS

# **People data** is personal

Workforce charts describe colleagues, and carry extra obligations.

### Minimum group size

No group under **10 people**: not in bars, labels, tooltips, filters or drill-downs. Merge into "Other" and footnote it.

### No re-identification

Small team × tenure × location can single out one person, even when each cut is large.

### Protected attributes and forecasts

Protected attributes only for approved analyses such as pay equity, aggregated. Label estimates and show a range. Frame findings about roles, not individuals.

**Engagement by team, one distribution center**

Two teams have fewer than 10 respondents: anyone who knows the team can guess who answered what.

| Team | Respondents | Engagement |
|---|---|---|
| Inbound | 46 | 71 |
| Outbound | 38 | 68 |
| Returns | 7 | 52 |
| Quality | 4 | 88 |
| Site leadership | 9 | 81 |

**Try to avoid**
Reporting groups below the minimum, even in a table or tooltip.

**Engagement by team, one distribution center**

Small teams combined into one group that clears the threshold.

| Team | Respondents | Engagement |
|---|---|---|
| Inbound | 46 | 71 |
| Outbound | 38 | 68 |
| Other teams (3 combined) | 20 | 72 |

**Recommended**
Merge into "Other" and footnote it: "Teams with fewer than 10 respondents are combined."

### In dashboards, enforce it in the data model

A rule in a slide is easy to forget; a rule in the measure can't be skipped. In Power BI: `IF([Headcount] < 10, BLANK(), [Engagement])`. In Python, drop or merge small groups before plotting, and keep individual values out of hover text.

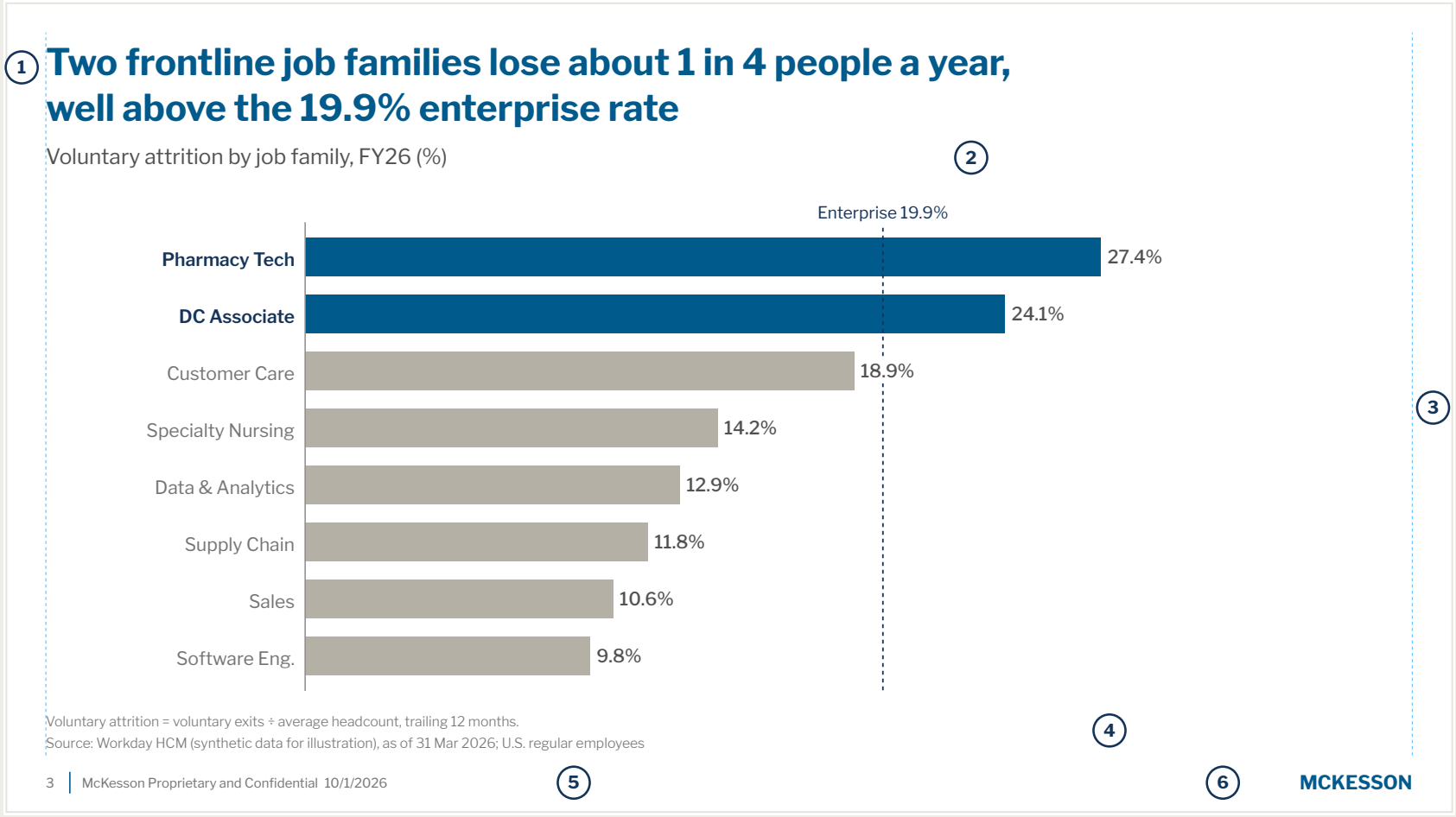
FORMATS

# The McKesson slide

Start from the McKesson Standard Template's *Title Only* layout: grid, footer and logo come with it.

1. **Insight title**
   Bold, McKesson Blue, 24–28 pt; two lines at most.
2. **Subtitle**
   Metric, unit, scope, period.
3. **Margins**
   All inside 0.39 in, one left edge.
4. **Source**
   Ends just above the footer.
5. **Footer**
   Page, confidentiality, date: leave on.
6. **Logo**
   From the master; never move it.

The orange underscore is for covers and section headers only: on chart slides, orange means "worse".



FORMATS

# **Presentations:** clarity first

A slide is on screen for less than a minute. There is no time to explore it.

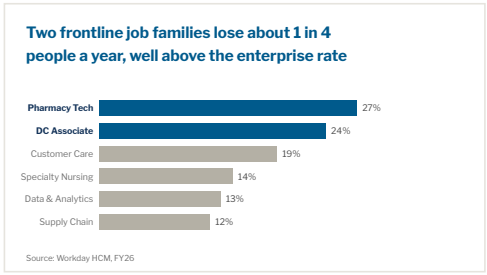
- With one chart, the slide title **is** its headline (Bold)
- One message per chart; at most two charts per slide, never a chart wall
- Short text beside the chart, or none
- Native, editable charts from the starter deck, not pictures of charts
- In a dimmed room, use the dark slide: navy background, light-blue focus

## Chart wall



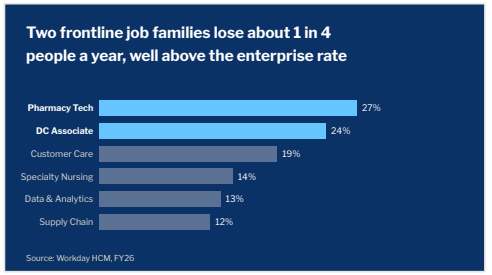

**Try to avoid**
A label title, four charts and a column of bullets.

## One message




**Recommended**
Headline as title, one chart, a source line.

## Dimmed room




**Recommended**
The same slide on navy; focus in light blue.

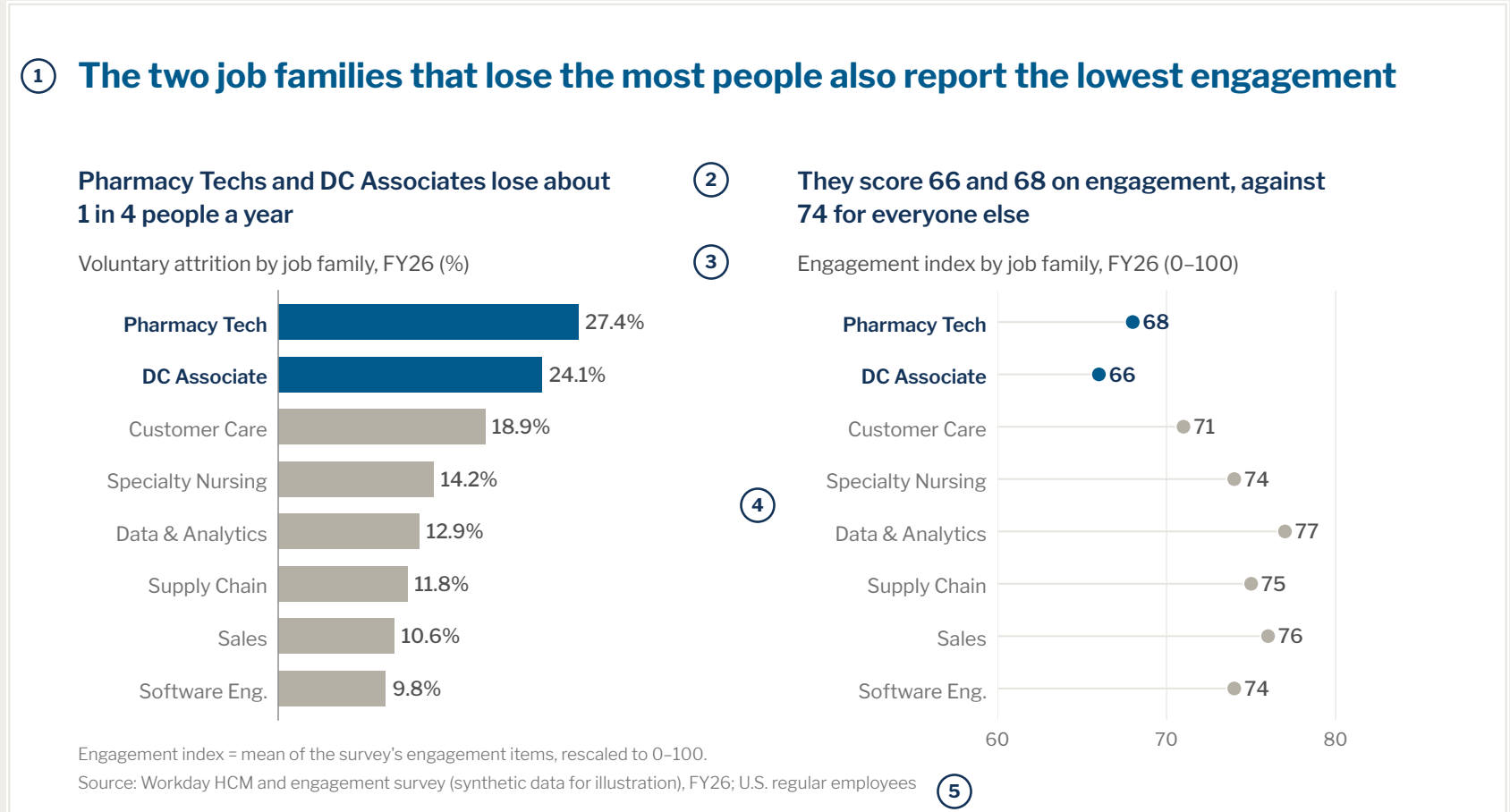
FORMATS

# **Two charts** on a slide

One chart per slide is the default. When the message needs two pieces of evidence, the titles form a hierarchy.

1. **Slide title, Bold**
   The one message both charts support.
2. **Chart headline, SemiBold**
   Each chart's own finding: still an insight, never a label like "Engagement by job family".
3. **Subtitle per chart**
   Metric, unit, scope and period, as always.
4. **Aligned and consistent**
   Headlines on one top line; same rows, same order, same highlight.
5. **One source line**
   Shared at the bottom when the charts share sources.

Two is the limit. A third chart means a second slide.



FORMATS

# **Flat,** not rounded

Rounded pills, soft shadows and gradients date a slide and make it feel less precise. Our look is flat and square.

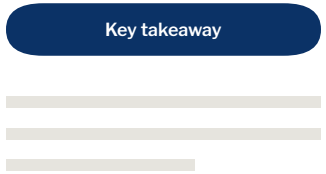
**Square corners.** Labels, panels and bars have square corners. No pills or capsules.

**No shadows or gradients.** Separate panels with white space, a sand fill or a thin rule.

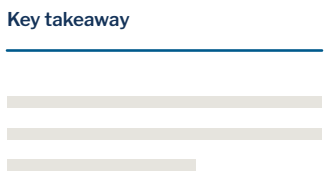
**Rules, not boxes.** A thin blue rule under a header does the work of a colored capsule.

**Words with status dots.** A colored dot never stands alone; the word sits beside it.

## Labels

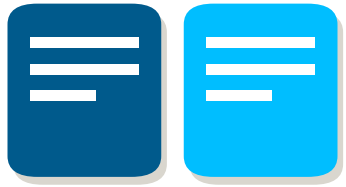



**Try to avoid**
A rounded pill.




**Recommended**
Bold label, thin rule.

## Panels

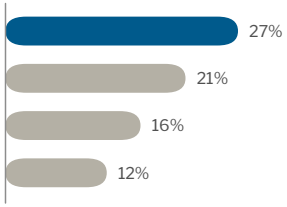


**Try to avoid**
Rounded, shadowed, bright.

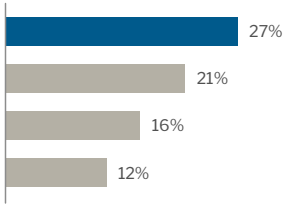


**Recommended**
Flat sand, a blue rule on top.

## Chart marks

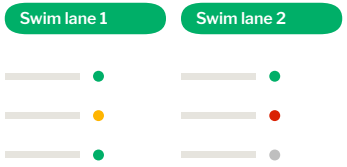



**Try to avoid**
Rounded bar ends.




**Recommended**
Square ends, true lengths.

## Trackers




**Try to avoid**
Pills; dots with no words.




**Recommended**
Plain headers; dot and word.

MAGNITUDE

# **Bar** chart

Magnitude charts compare size: which is bigger (relative), and by how much (absolute).

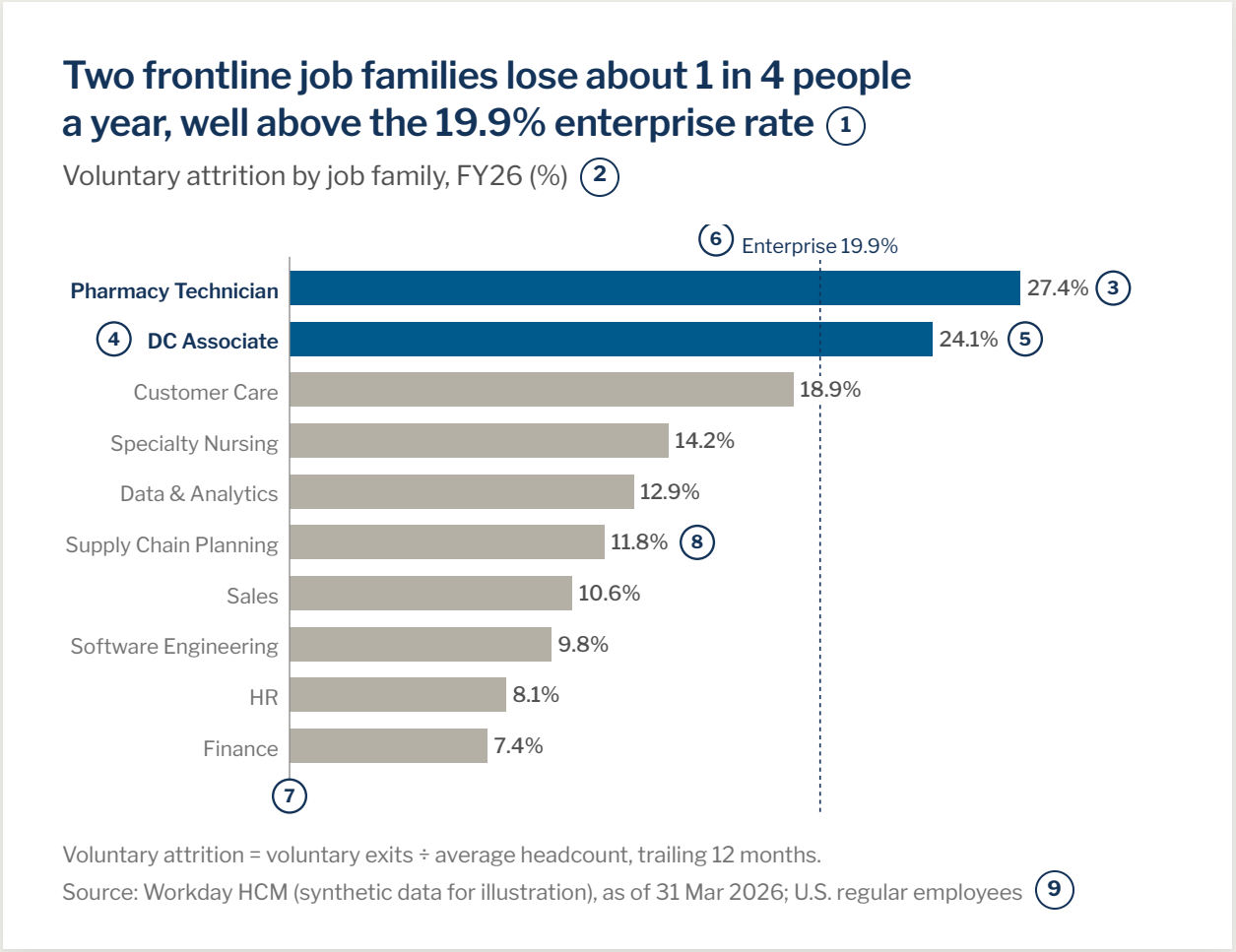
A bar's length is proportional to its value, so bars are the most precise way to compare categories. Sorted horizontal bars are the default for ranking job families, sites or skills: long names stay readable, and the layout works on a phone.

Use columns for a few time periods or short names.

### McKesson spec

- Bar about 2× the gap (Excel gap width 50%)
- Sorted by value unless there's a natural order
- Up to about 12 bars; group the tail into "Other"
- No group under 10 people

1. **Insight headline**
   Names the two families and the benchmark they beat.
2. **Subtitle**
   Metric, unit, scope, period.
3. **Focus in blue**
   Only the bars the headline names; two at most.
4. **Echo in the label**
   Focus categories in bold navy.
5. **Values on the bars**
   So the value axis and gridlines go.
6. **Labelled benchmark**
   A thin dashed line, named where it sits.
7. **Zero baseline**
   A thin axis line; bars always start at zero.
8. **Grey context**
   Everything else, sorted by value.
9. **Footnote and source**
   Defines the metric; names system, date, population.



MAGNITUDE

# **Bar chart** best practice

Four rules that fix most bar charts.

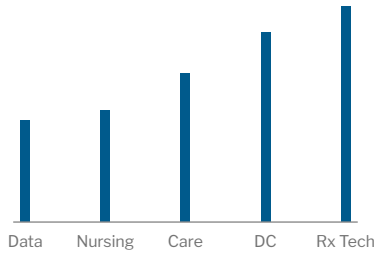
**Size of bars.** Thin bars pull the eye to the empty space between them. Make each bar about twice as wide as the gap.

**Color of bars.** A different color per bar distracts and implies categories that don't exist. One color; shade only the highlight.

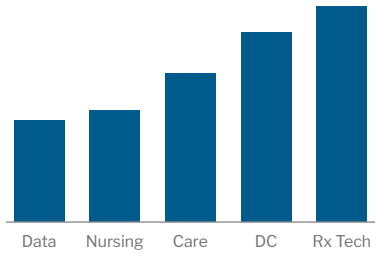
**Direct labels.** Gridlines and a value axis add noise. Label the bars and delete both.

**Zero baseline.** Bar length encodes quantity, so never truncate. If the data needs a narrow range, switch to a line or dot plot.

## Size of bars

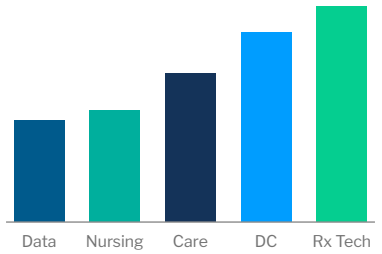



**Try to avoid**
Stick-thin bars; the gaps dominate.

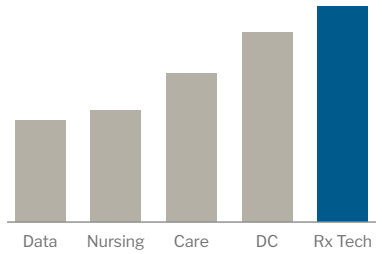



**Recommended**
Bar about twice the gap.

## Color of bars

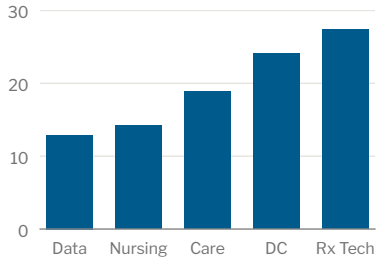



**Try to avoid**
A color per bar, encoding nothing.

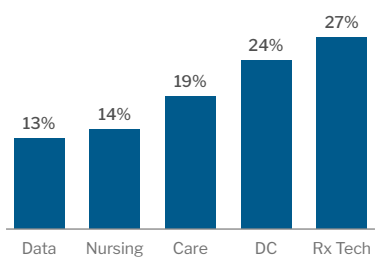



**Recommended**
Grey bars; blue for the one that matters.

## Direct labels

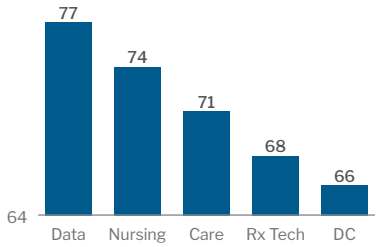



**Try to avoid**
Gridlines and an axis to read values from.

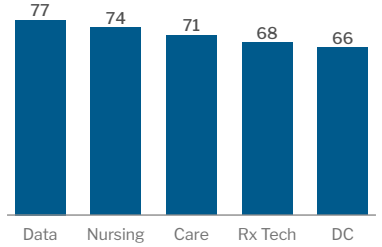



**Recommended**
Values on the bars; no axis, no grid.

## Zero baseline




**Try to avoid**
Cut at 64, 77 looks 6.5× 66.




**Recommended**
From zero: 17% higher, as it is.

CHANGE OVER TIME

# **Line** chart

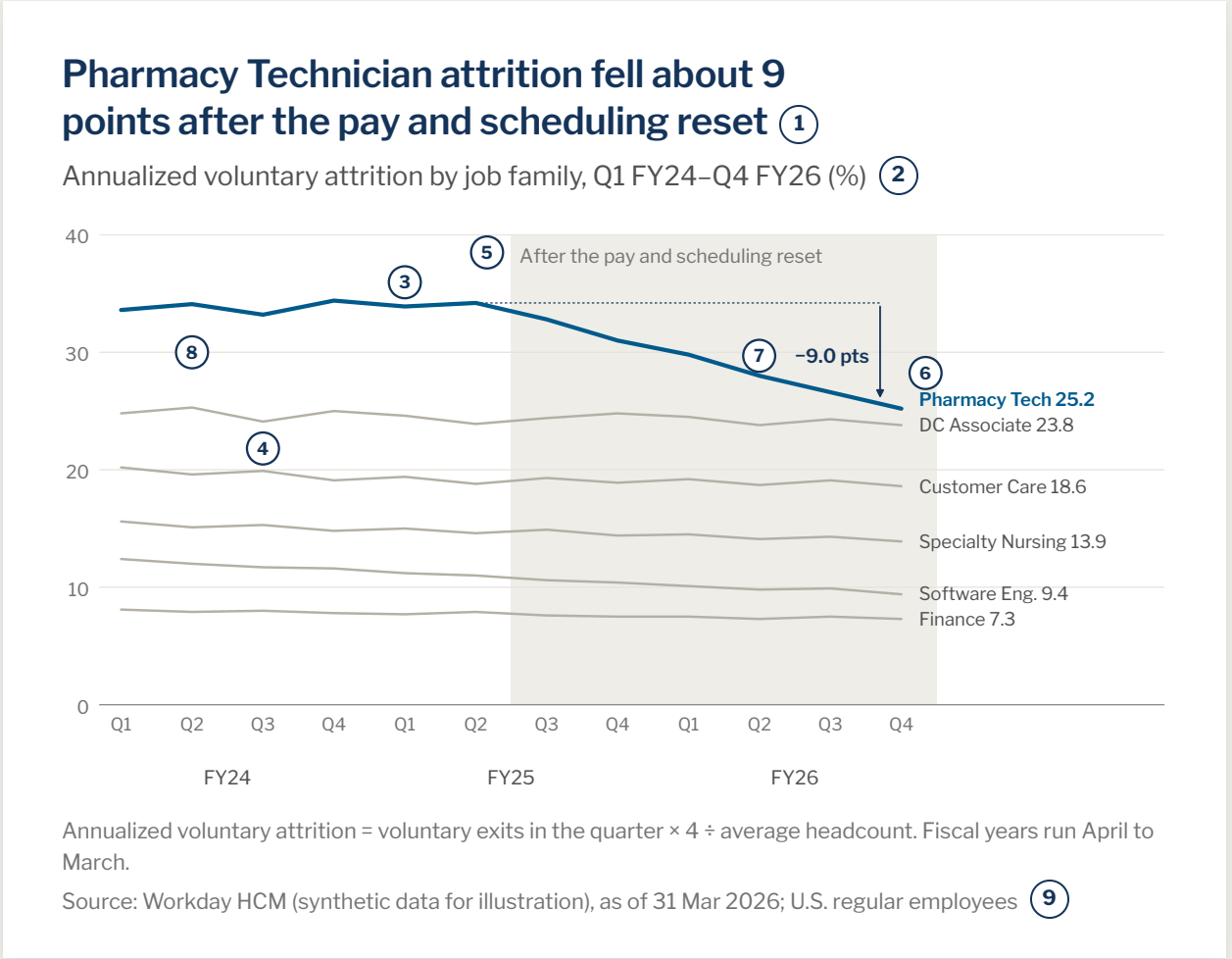
Change-over-time charts show how something moves, and when it changed, at any time scale from weeks to decades.

A line connects values at equal time intervals, so the eye reads the trend rather than the individual amounts. Use it for monthly or quarterly attrition, hiring velocity, or a forecast.

### McKesson spec

- The series the headline names in blue; others grey
- Data lines about twice the gridline weight
- Names at the line ends, not in a legend
- Shade the period that explains the change

1. **Insight headline**
   What changed, by how much, and why.
2. **Subtitle**
   Metric, unit, scope and period.
3. **Focus line**
   Blue and heavier than the rest.
4. **Context lines**
   Grey and thinner: there to compare against.
5. **The period that matters**
   Lightly shaded and named.
6. **Direct labels**
   Names and latest values at the line ends, nudged apart.
7. **The change, sized**
   A short callout carries the headline number.
8. **Light scaffolding**
   A few thin gridlines; the unit lives in the subtitle.
9. **Footnote and source**
   Define the metric.



CHEAT SHEET

# **Cheat** sheet

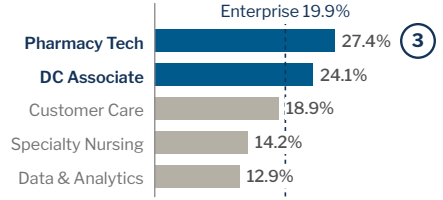
## Check before you send

- [ ] The headline is an insight, and every number in it is true
- [ ] Subtitle: metric, unit, scope and period
- [ ] Grey context; at most 2 highlights, matching the headline
- [ ] Values labelled; value axis, gridlines and legend deleted
- [ ] Bars start at zero; no dual axes
- [ ] Sorted by value, or in a natural order
- [ ] True minus (−), explicit + on changes, "pts" for point changes
- [ ] Metric defined; source names system, date and population
- [ ] No group under **10 people**, even in hover text
- [ ] Nothing overlaps or clips; nothing below 9 pt

## 1 · EVERY CHART, ONE ANATOMY

**Two frontline job families lose about 1 in 4 people a year, well above the 19.9% enterprise rate** (1)

Voluntary attrition by job family, FY26 (%) (2)




Source: Workday HCM (synthetic), FY26 (4)

(1) Headline = the insight · (2) Subtitle = the measure · (3) Grey context, blue focus, values on the bars · (4) Source

## 2 · WRITE THE HEADLINE LAST

Who or what + verb + number + so what. Compute every number from the data.

| | |
|---|---|
| **LABEL** | Attrition by job family |
| **VAGUE** | Attrition varies across job families |
| **INSIGHT** | Two frontline job families lose about 1 in 4 people a year |

## 3 · GREY FIRST, THEN COLOR THE POINT

| | |
|---|---|
| **Context** everything else | #B4B0A5 |
| **Focus** what the headline names | #005A8C |
| **Accent (navy)** the one exception | #143359 |
| **Light blue** a series, in the uniform pattern | #66C4FF |
| **Orange** worse only: below target, shrinking | #EF8200 |
| **Green** better only, with ▲ or words | #00AF68 |

At most 2 highlights. Orange and green only mean worse and better. Red, amber, green: status only, with a symbol and word.

## 4 · TYPE (LIBRE FRANKLIN, 16:9 SLIDE)

| | | |
|---|---|---|
| **Slide title** | Bold | 26 pt |
| **Chart headline** | SemiBold | 18 pt |
| **Subtitle** | Regular | 14 pt |
| **Data labels** | Medium | 12 pt |
| **Axis, categories** | Regular | 11 pt |
| **Footnote, source** | Light | 9 pt |
| **Big number** | Bold | 54 pt |

One chart: the slide title is its headline. Two: each its own SemiBold insight headline.

## 5 · PICK THE CHART FROM THE QUESTION

- How big? Sorted bar
- Made of what? 100% bar, waterfall
- Over time? Line
- Who moved? Slope, dumbbell
- How spread? Strip, histogram
- Related? Scatter
- Flows? Sankey · Where? Map
- One number? Big number

Full version: the chart chooser.

## 6 · NEVER

- Red vs. green, rainbows, or color as the only cue
- 3D, shadows, gradients, borders
- Truncated bars or dual axes
- A legend where labels fit
- More than 2 charts on a slide
- Pictures of charts: use native, editable charts

TOOLKIT

# **Set up** once

For decks and workbooks, most people need to install nothing: start from the template and copy charts from the starter deck.

Everything named here is in the kit: the release bundle (a zip on SharePoint) or the GitHub repo. More detail and troubleshooting:

- docs/office-setup.md
- docs/powerbi-setup.md
- AGENTS.md (Python, HTML, R)

| | |
|---|---|
| **Check the font** | In PowerPoint, type **Libre Franklin** in the font box. If it's listed, you're set. If not, open the kit's **fonts** folder, select every file, right-click › **Install** (or ask IT). |
| **PowerPoint** | **New deck:** File › New › the **McKesson Standard Template**. **Charts:** open the kit's **starter deck**, copy a chart slide into your deck, then right-click the chart › **Edit Data** and paste your numbers. Colors, fonts and layout come with the slide. |
| **Excel** | Open the kit's **chart workbook**: replace a sheet's data, or copy its chart into your file. The **Palette** sheet lists every color with the hex to type into More Colors. Want the colors in your own workbooks? The optional theme files are in office-setup. |
| **Power BI** | **View › Themes › Browse for themes** and pick `dist/powerbi/mckesson-dataviz-theme.json`. Viewers without Libre Franklin see Segoe UI, so leave room in labels. |
| **Python** | `$kit = "git+https://github.com/amccown-mck/Data-Viz-Guide@v1.1.0#subdirectory=python"`<br>`pip install "mckviz @ $kit"  # fonts and maps included`<br>`import mckviz as mv`<br>`mv.use()  # fonts, matplotlib style, Plotly template "mck"`<br>`C = mv.colors  # C.focus, C.context, C.accent, C.negative ...` |
| **HTML** | Link `dist/web/mck-viz.css` for the CSS variables and chart-card styles; add `mck-plotly-template.js` for Plotly or use `vega-lite-config.json` as the Vega-Lite `config`. Start from `templates/html/chart-card.html`. |
| **R** | `source("dist/r/theme_mck.R")`, then `ff <- mck_font()` and `theme_mck(base_family = ff)`. Render with `ragg`. |